Grade: Check Plus

STERN SCHOOL OF BUSINESS
NEW YORK UNIVERSITY

EXECUTIVE MBA
DC CLASS OF 2027

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Statistics and Data Analysis

ASSIGNMENT 2

Course Supplement, Page 129:

EXPECTED VALUES AND OPTIONS

Due: February 24, 2026 before 6 PM

1. EXPECTED VALUES AND OPTIONS

A stock has a current selling price of $20. The distribution of the price of the stock one year from now is believed to be as follows:

PRICE (x)	PROBABILITY f(x)
20	.05
21	.06
22	.07
23	.08
24	.09
25	.12
26	.15
27	.13
28	.10
29	.10
30	.05
	1.00

An investor holds a one year option (European) on this stock. This enables him to purchase, one year from now, one share of stock at $25 per share. The value of his option will depend upon the price of the stock one year from now. If the price rises to $28, the value of the option (ignoring interest and commissions) is $3. If the price rises to only $23, the option is worthless.

1. Compute the mean and standard deviation of the stock price.
2. a. What is the probability that the stock price will fall within 1 standard deviation of the mean one year from now?
   b. What is the probability that the stock price will fall within 2 standard deviations of the mean one year from now?
3. Derive the distribution of the option value, ignoring interest and commissions.
4. What is the probability that the option will have positive value?
5. Compute the mean and standard deviation of the value of the option.

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1. Compute the mean and standard deviation of the stock price.

Mean:

20*0.05+21*0.06+22*0.07+23*0.08+24*0.09+25*.12+26*.15+27*.13
+28*.1+29*.1+30*0.05

===> 25.41

The weighted average (mean) of the stock price is 25.41, which is $25.41

Standard deviation:

We calculate the standard deviation first by using **$x^2 \cdot f(x)$** which for each of the values is as follows:

20.00+26.46+33.88+42.32+51.84+75.00+101.40+94.77+78.40+84.10+
45.00

= 653.17

Then we use this formula σ=√E(X2)−(μ)2 to compute the variance:

= 653.17-25.41^2
= 653.17-645.67
Variance = 7.50

σ=√7.50

✓ Q1

===> σ=2.74

2(a) the probability that the stock price will fall within 1 standard deviation of the mean one year from now?

σ=2.74

Lower bound: 25.41-2.74 = 22.67
Upper bound: 25.41+2.74 = 28.15

So we want P(22.67$\leq$X$\leq$28.15)

X = 23, 24, 25, 26, 27, 28

P = 0.08+0.09+0.12+0.15+0.13+0.1 = 0.67

✓ Q2a

===> P (within one Standard Deviation) = 0.67

2(b) What is the probability that the stock price will fall within 2 standard deviations of the mean one year from now?

σ=2.74

Lower bound: 25.41-(2)*2.74 = 19.93
Upper bound: 25.41+(2)*2.74 = 30.89

So we want P(19.93$\leq$X$\leq$30.89)

X = 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30

X = every possible stock price in the table
X = 0.05+0.06+0.07+0.08+0.09+0.12+0.15+.13+.1+.1+0.05 = 1

===> P (within two Standard Deviations) = 1.00

✓ Q2b

3. Derive the distribution of the option value, ignoring interest and commissions.

We are told the option is a European call with a price of 25, so the option value one year from now is: $V=\max(X-25,0)$

Part 1: Map stock price x to option value v = max(x-25,0)
If $x \leq 25$, then v = 0
If x=26, v=1
If x=27, v=2
If x=28, v=3
If x=29, v=4
If x=30, v=5

Part 2: Option value v = 0 happens when x = 20,21,22,23,24,25
P(V=0)=0.05+0.06+0.07+0.08+0.09+.12=0.47

Part 3:

Option Value (v)	**Probability (P(V=v))**
0	0.47
1	0.15
2	0.13
3	0.10
4	0.10
5	0.05
Total	**1.00**

✓ Q3

4. What is the probability that the option will have positive value?

Using the table from above:

P(V>0)=0.15+0.13+0.1+0.1+0.05=0.53

✓ Q4

===> The probability that the option will have a positive value is 0.53.

5. Compute the mean and standard deviation of the value of the option.

v (option value)	P(V = v)
0	0.47
1	0.15
2	0.13
3	0.10
4	0.10
5	0.05

Step 1: Build the table:

v	P(v)	v·P(v)	$v^2 \cdot P(v)$
0	0.47	0.00	0.00

1	0.15	0.15	0.15
2	0.13	0.26	0.52
3	0.10	0.30	0.90
4	0.10	0.40	1.60
5	0.05	0.25	1.25
Total	**1.00**	**1.36**	**4.42**

Mean = $E[X]=\sum xf(x)$

===> Mean = 1.36, or $1.36

Standard deviation = $\sigma=\sqrt{E(X2)-(\mu)2}$

$=4.42-1.36^2$
$=4.42-1.85$
Variance = 2.57

$\sigma=\sqrt{2.57}$

✓ Q5

===> σ=1.60, or $1.60